AMANDA SEWWANDI

Computer Science Undergraduate | Software, Data and AI
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PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate (GPA 3.58) with strong programming fundamentals in Python and JavaScript, solid SQL and relational database skills, and hands-on experience across full-stack development, machine learning, MLOps and retrieval-augmented generation (RAG). Built and deployed REST APIs, database-backed applications with transactional integrity, and AI systems using Hugging Face and LLM APIs. Wrote automated tests and took part in code review on team projects. Worked with real banking records as an Intern Bank Assistant at the Bank of Ceylon, where accuracy and compliance were essential. Clear written and spoken English.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL, R, NetLogo; data structures and algorithms; REST API development (Node.js/Express, FastAPI)
AI and Machine Learning: Retrieval-Augmented Generation (RAG), LLM APIs (Groq Cloud), Hugging Face (Datasets, Transformers), PyTorch, TensorFlow/Keras, Scikit-learn, SHAP/LIME, Optuna, MLflow
Databases & SQL: SQL queries, relational schema design, transactions (commit/rollback), MySQL, PostgreSQL, SQL Server, SQLite, Prisma ORM
Data & Visualization: Pandas, NumPy, SciPy, Matplotlib, Plotly, Streamlit, Tableau Public
Testing, DevOps & Version Control: Pytest, Docker and Docker Compose, CI/CD with GitHub Actions, Git/GitHub (branches, pull requests, code review), Azure deployment basics
Foundations: Data Structures and Algorithms, Database Systems, Software Engineering (SDLC, Agile/Scrum), Statistics, Machine Learning, Agent-Based Modelling
Languages: English (written and spoken), Sinhala, Tamil

PROJECTS

ERP System, University of Peradeniya | Project Manager | 2026 – Present
- Currently serving as Project Manager for an ERP project at the University of Peradeniya, leading the team and managing planning and delivery.

RAG System: Question Answering with Retrieval-Augmented Generation | 2026
- Built a retrieval-augmented generation system that answers questions using a dataset sourced from Hugging Face, grounding responses in retrieved context to reduce hallucination.
- Connected a large language model through the Groq Cloud API for fast inference, combining retrieved passages with the user's question in a structured prompt.
- *Tech: Python, Hugging Face Datasets, Groq Cloud API, RAG*

Firefighter Paradox: NetLogo Wildfire Agent-Based Model | 2026
- Built an agent-based wildfire model in NetLogo 7.0.4 with a forest, village, firefighters, helicopters and villagers, where fire spreads faster uphill and downwind, is slowed by humidity and cannot cross rivers; 11 adjustable parameters include tree density, wind, humidity, lightning rate, budget and crew size.
- Ran BehaviorSpace experiments to study a tipping point and the firefighter paradox: below about 40% tree density fires stayed small (5–15% burned), while above 60% about 97–99% of the forest burned; adding more firefighters and helicopters gave no clear drop in the biggest fire.
- *Tech: NetLogo 7.0.4, BehaviorSpace, agent-based modelling, simulation, CSV/Excel result analysis*

Brew & Bite: E-Cafe Management System | Individual Project | 2026
- Designed and built a full-stack cafe management system covering menu, customers and orders with status tracking, using a Node.js/Express REST API (15 endpoints) and a JavaScript frontend.
- Modeled the MySQL schema and implemented order placement as a database transaction (commit/rollback) to protect data integrity; prepared Azure deployment with environment-variable secrets, SSL connections and a GitHub Actions CI/CD workflow.
- *Tech: Node.js, Express, MySQL, REST APIs, JavaScript, Azure, GitHub Actions*

KLOSET: Multi-Vendor Fashion E-Commerce Platform | University Group Project | 2025
- Contributed to a team-built marketplace (210+ commit repository): React, Redux and Tailwind CSS frontend and a Node.js/Express REST API with Prisma ORM on PostgreSQL (Supabase), including user authentication and a vendor dashboard.
- Collaborated through Git branches, pull requests and code review discussions to merge team contributions.
- *Tech: React, Redux, Tailwind CSS, Node.js, Express, Prisma, PostgreSQL, Supabase, Git*

Student Depression Predictor: Dockerized MLOps Pipeline | 2025
- Built a reproducible training and evaluation pipeline with Docker Compose and automated Pytest tests in a CI/CD workflow (128+ commits).
- *Tech: Python, Scikit-learn, Docker, Pytest, CI/CD*

FinSight Lanka: Customer Data Analysis and KPI Dashboard | Case Study
- Audited and cleaned a 508-row customer dataset (500 after cleaning): resolved duplicates using evidence, standardised districts and provinces, recalculated ages, flagged contradictions and kept a cleaning log with validation checks.
- Analysed savings, loan risk, cross-sell readiness and digital engagement using regression and bootstrap stability tests; built a Tableau Public KPI dashboard tracking five monthly KPIs against targets (loan delinquency, KYC verified, active account, cross-sell coverage, fragile saver rates).
- *Tech: Python, Pandas, NumPy, SciPy, Matplotlib, Jupyter, Tableau Public*

BBC News MLOps Pipeline: Automated News Classifier | 2026
- Built and containerized a news-classification model (98.12% accuracy) served through a FastAPI endpoint, with automated retraining and experiment tracking in MLflow.
- *Tech: Python, PyTorch, Docker, FastAPI, MLflow*

Multimodal Churn Intelligence Framework (MCIF) | Coursework Research Paper | 2026
- Built an explainable ML pipeline predicting customer churn on a telecom dataset, improving recall on high-value churners with ensemble classifiers and Optuna-tuned hyperparameters.
- Applied SHAP/LIME and cost-sensitive threshold optimization to align predictions with business costs; deployed a Streamlit app so non-technical stakeholders can explore churn drivers.
- *Tech: Python, Scikit-learn, SHAP/LIME, Optuna, Streamlit*

IMDB Sentiment Analysis: Fine-Tuned DistilBERT Classifier | 2026
- Fine-tuned a DistilBERT transformer with Hugging Face for text classification and shipped a live Streamlit app for real-time predictions.
- *Tech: Python, PyTorch, Hugging Face Transformers, Streamlit*

Immune Landscape of Tumors: Data Visualization and Dashboard | CFDE Competition | 2026
- Designed an interactive dashboard that turns multi-cancer clinical data into clear, stakeholder-friendly insights on tumor composition, mutation burden and survival trends.
- *Tech: Python, Streamlit, Plotly, Pandas, NumPy*

Traffic Sign Classification: CNN Deployment | 2025
- Developed a CNN classifying 43 traffic sign types at 97.70% test accuracy and deployed it as a live Streamlit demo.
- *Tech: Python, TensorFlow, Keras, OpenCV, Streamlit*

EXPERIENCE

Intern Bank Assistant, Bank of Ceylon, Malabe | 10/2022 – 04/2023
- Processed and validated account-opening and transaction documents for 30+ customers daily with full accuracy, following compliance and customer-confidentiality requirements.
- Identified recurring data-entry inconsistencies in records and supported corrective actions, which started my focus on data quality.

Campus Coordinator, IEEE University of Sri Jayewardenepura | 2025
- Led cross-functional teams and coordinated technical workshops with students, academics and external partners, handling planning, scheduling and event documentation, and acting as liaison to IEEE regional bodies.

Volunteer Mathematics Teacher, Sasnaka Sansada | 2022 – Present
- Mentor underprivileged students through hands-on teaching, building communication and patience.

EDUCATION

Bachelor of Computing (Hons) in Computer Science, University of Sri Jayewardenepura, Colombo | 2023 – Present
- GPA 3.58 (through Year 3, Semester 1). Coursework: Data Structures and Algorithms, Database Systems, Software Engineering, Machine Learning, Statistics.

Diploma in English (Level 3, with distinction), Aquinas College of Higher Education | 2022

G.C.E. Advanced Level, Anula Vidyalaya, Nugegoda | 2022 – Combined Mathematics (A), Physics (C), ICT (B)

RESEARCH, PUBLICATIONS AND CERTIFICATIONS

Microsoft Certified: Fabric Data Engineer Associate (DP-700) | In progress, expected 2026

Task-Aware Quantization of Vision-Language Models | Final Year Research | 2025 – Present
- Benchmarking accuracy, latency, memory and energy trade-offs across six quantization configurations for real-time inference on edge devices.

Comparative Review: CycleGAN vs. Diffusion Models for Plant Disease Detection | 2026
- Published at the First Jayewardenepura Computing Symposium.

"Big Data: Powering the Digital Economy" | Association of Computing Students, USJ | 2025
- Published article explaining big data to a general student audience.

REFEREES

Dr. S. Majuran, Senior Lecturer, Faculty of Computing, University of Sri Jayewardenepura | shajini@sjp.ac.lk | 074 057 2678
Kanishka Gayan, Research Assistant and Visiting Lecturer, Dept. of Knowledge Engineering and Communication, USJ | kanishkaedirimanne@sjp.ac.lk | 071-6632537